

Syed Abdul Aleem

Lead Data Scientist | GenAI | Machine Learning | Agile Methodologies | Design Thinking

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PROFESSIONAL SUMMARY

13+ years of progressive experience in TCS & Happiest Minds Technologies with the last 7+ years in designing and developing models for regression, classification, NLP, and time series projects. Proven track record of success in deploying models into production with CI/CD pipelines and creating automated processes for data analysis. Skilled at creating insights from data, visualizing results, and developing innovative solutions for complex problems. Worked with large datasets, advanced statistical methods, and machine learning algorithms. 2+ years working in GenAI projects using Azure OpenAI & Langchain. Familiar with AI Agents and Finetuning LLMs

Techno-functional Leadership

Designed and developed an end-to-end data science solution utilizing Machine Learning concepts & algorithms, MLOps, strong team motivation and effective communication skills, resulting in a successful project within budget. Accomplished 10+ ML projects and 3 Gen AI projects in the last 5 years.

Growth Facilitator

Successful in developing and transitioning data science Proof Of Concept (POC) into production, overcoming challenges in workflows, data analysis, and ML frameworks. By leveraging facts, data, and context, made informed and rational decisions in terms of data, features, and algorithms for model training

VALUES OFFERED

- Demonstrated expertise in pre-processing, model deployment, and MLOps, and a strong understanding of programming languages, databases, and cloud platforms
- Led a successful POC to production project in March 2023, receiving client appreciation
- Built a team of 12 data scientists with varied skillsets and created a culture of learning and excellence
- Engaged with SMEs to understand business requirements and proposed the best tools and approach for the project in terms of cost and effectiveness
- Identified potential risks and proposed solutions to mitigate them
- Took initiatives in Gen AI POCs and successfully converted into a project
- Proposed architecture for Gen AI solutions based on the budget & requirements of client



EDUCATION & PROFESSIONAL DEVELOPMENT

2018 PGPM Executive MBA in Data Analytics – Great Lakes Institute of Management

2011 B Tech in Mechanical Engineering – SRM University

AZ-900: Microsoft Azure Fundamentals Certification

Training on Data Science with Python – Learnbay

Training on MLOps Fundamentals, Time Series Analysis, Forecasting, and Machine Learning – Udemy



TECH SKILLS

Machine Learning Concepts	Machine Learning Algorithms	Programming
Statistics Descriptive & Inferential Statistics Hypothesis Testing & Probability Learning Libraries Scikit-Learn, Numpy, Pandas, Pyodbc Matplotlib, Seaborn, SQL alchemy, Spacy, NLTK, OpenCV & TensorFlow Pre-Processing Data Cleaning, Data Transformation, Feature Engineering, PCA, Hyperparameter Tuning, Data Normalization, Label Encoding, Tokenization. Parts of speech tagging,	Regression Linear Regression, Decision Trees, Random Forest, XGBoost Regressor, SVM, Multiclass Regressor Classification LRT, Decision Trees, Random Forest, XG Boost Classifier, Naive Bayes, Multiclass Classifier NLP TF-IDF, Bag of Words, Summary Extraction, Text Classification, Key Phrase Extraction, Fast Text Jaccard Similarity Time Series Forecasting XGBoost, Arima, Moving Average, Auto Regression	Languages Python, R , SQL, MS Excel Database SQL, MongoDB, NoSql, Cloud Platform, Azure, IDE, Jupyter Notebook, Spyder Visual Studio Code, Azure Databricks Repos, Bitbucket, GitHub Model Deployment

Named Entity Recognition

MLOps

Create CI/CD Pipelines and reproducible environments, Process Model versioning & monitoring, and evaluate model performance

Time Series Anomaly Detection

Dynamic Thresholding, Isolation Forest

Image Classification

CNN, Morphological Operations, Contour Identification

Gen AI

Azure OpenAI, Langchain, Prompt Engineering, Chain of Thoughts, AI Agents

Rest APIs, Azure VMs, Docker

Images,

Azure Machine learning,

Workspace

Kubernetes, Azure Blob Storage



EXPERIENCE AND MILESTONES

HAPPIEST MINDS TECHNOLOGIES ▶ Lead Data Scientist

Nov 2023 – Present

PoC Design & Development for Gen AI | Build architecture for Gen AI solutions | Resource planning based on client budget

KEY PROJECTS EXECUTED

GEN-AI

Sentiment Analysis & Call Reasons – Analysed 2 years of daily call transcripts between different customers & call-center agents using Gen AI. Based on the analysis, obtained the sentiment of customer during the call and grouped it into Satisfied/Unsatisfied/Angry. Reason for which the call was made was also identified. The results for each conversation id was stored in a SQL table which in turn was used in the building Power Bi Dashboards to give meaningful insights. This was a POC which was approved as a project. Proposed the architecture for the solution and it was successfully deployed into production.

Player Summary – For an ongoing sport league, web scrape most recent information based on form, availability & injury and provide a summary for each player which would help the users to make an informed decision to build their teams in betting apps.

This was a POC which was successfully converted into a project. Proposed and presented the architecture for this solution and the RFP.

Product Recommendation – Based on user requirements in terms of utility, budget & preferences the gen ai app will web scrape the latest information available on similar products and recommend the best two products for the customers.

This was a POC that helped me receive an award in the organization.

TIME SERIES FORECASTING

Forecasting Sales of Beer in US and Canada – Built a timeseries forecasting model using Prophet and Log-Log method to forecast the sales of Beer in US and Canada at every month. Identified the key regressors which is driving the growth / decline and the impact it had on the forecasting.

TATA CONSULTANCY SERVICES ▶ Data Scientist

Jan 2018 – Oct 2023

PoC Design & Development | Proof of Concept to Production | Develop end-to-end data science solutions |
Led 12+ data scientists | SME Collaboration | Suggested cost & process efficient tools & approaches

KEY PROJECTS EXECUTED

REGRESSION

Digital Calibration Frequency - Designed, developed and deployed a linear regression model for predicting the number of days for the next calibration of devices, using **Weibull Analysis and Azure Databricks. Automated CI/CD pipelines** to handle changes without manual intervention

Predicting Concentration Value - Predicted the concentration value for a spectrum dataset using **PCA and SVM Regressor**. Principal Component Analysis was used to reduce the dimensionality of the dataset and replicate the functionalities of an existing tool including the Hotelling T2 plots. The model achieved 100% accuracy in matching the predictions with the existing tool.

CLASSIFICATION

Product Classification - Developed a **multi-class logistic regression** model to predict the category of products based on input values. Deployed the model as a pickle file into **Azure Blob**, and created an **API** to predict the results each time a new file is uploaded. Also created **CI/CD PIPELINES** TO HANDLE CHANGES WITHOUT MANUAL INTERVENTION.

NATURAL LANGUAGE PROCESSING

Data Linkage – Phrase Extraction - Led efforts in the extraction of Site Information, Location, and Address details from available text information using **Fuzzy Matching and Levenshtein distance algorithms**. **Deployed the model** into an in-house platform, and the project was approved for the next phase

RSC - Developed a proof-of-concept that used **FastText and Jaccard Similarity** to generate a maximum similarity score between two addresses above the threshold limit and assign the address to the predictive field. Applied **standard text cleaning techniques** to remove special characters and punctuation. Project assigned to Physical Manufacturer.

Insights - Key Phrase Extraction & Summary - Developed and deployed a model for key phrase extraction and summarizing customer reviews and complaints. Used **TF-IDF and Word & Sentence Tokenization** to clean and analyze the text, and generated summaries based on a threshold limit. **Deployed Model** in an in-house platform and the output were very useful to the end user

TIME SERIES

Forecasting OEE - Developed a forecasting model for **Overall Equipment Effectiveness (OEE) of different machines using XGBoost**. Used key fields that decide the OEE value of the machines and the data was stored in a database. The training and prediction codes were deployed into an **Azure Virtual Machine** and a **cron-job** was created to run the codes as per the requirements. The model was saved into **Azure Blob Storage** and also created CI/CD pipelines for deployment

Anomaly Detection of Sensors – Led anomaly detection of sensors by using static thresholding. Analyzed **IOT data** received from sensors and used it to forecast anomalies. Fixed a threshold limit of temperature and classified those points as anomalies.

IMAGE CLASSIFICATION

Contact Lens Defect Identification - Developed an automated defect recognition tool for contact lenses using a **Convolutional Neural Network (CNN) with RELU activation**. The images were converted to black and white using **adaptive thresholding and morphological operations** such as MORPH_RECT, erode and dilate were used to detect any lines in the contact lens images. The model achieved an accuracy of 98%, reducing costs and manual effort by 70%

TATA CONSULTANCY SERVICES ▶ Data Engineer

Aug 2016 – Dec 2017

Active participation in Sprint Review Meetings | Tableau Reporting | Data Transformation | Knowledge Sharing

- Delivered projects successfully in tight deadlines for Johnson & Johnson, a leading client in the Life Science domain
- Handled large volumes of data using **Azure Databricks**, and used tools such as **SQL, MySQL, Python**
- Developed pipeline & stages for data transformation from raw source system data into readable columns & storing in SQL DB
- Extracted data from different source systems and combined them into common tables.
- **Contributed to several projects**, such as ASPAC Consumer Japan Tableau Connection, IDiscover Hospital Reports, and One MD Change Management
- **Roles and responsibilities** included active participation in scrum calls, sprint reviews and meetings, providing detailed reports on the scenarios executed, creating Tableau reports for different views based on requirements, and analyzing scenarios for testing

TATA CONSULTANCY SERVICES ▶ IT Analyst

Jan 2015 – Jul 2016

Cross-functional Team Collaboration | Continuous Process Improvement | Change Management

- Managed end-to-end contracts and invoicing activities for TCS, incorporating resource work orders, project work orders, and support work orders
- Developed dashboards to identify non-compliant engagements at the time of project initiation
- Generated portfolio-wise revenue summary every quarter and identify trends and patterns
- Led change management initiatives in the contracts & invoices space of TCS-J&J and brought about changes in the operations processes

- Devised strategies for the invoicing process and led discussions with the Operations Head on the planning of Quarterly/Annual revenue.
- Groomed the team for improved performance

TATA CONSULTANCY SERVICES ▶ PMO Lead**2012 April – 2014 December****Project management and Delivery | Dashboard Creation | Resource Management**

- Completed the invoicing in the internal finance system within the deadlines for Johnson & Johnson (J&J)
- Reconciled customer invoices with the internal finance team to ensure the accuracy of data
- Created dashboards to identify non-compliant engagements at the time of project initiation
- Worked closely with TCS Resource Management Group to fulfill requirements of ongoing projects
- Monitored resource utilization against bench capacity to optimize resource usage

**ADDITIONAL INFORMATION****Major involvements**

- Led TCS Maitree ITPL Toastmasters Club as President, increasing candidate engagement
- Spearheaded CSR initiative to teach basic English to government school kids
- Implemented successful employee engagement initiatives, resulting in a 20% increase in employee satisfaction

Date of Birth – 16 March 1990**Languages Known** - English, Hindi & Tamil